

A clinical trial comparing the occlusal parameters and wear of artificial teeth between lingualized and anatomical occlusal complete dentures

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Abstract

Object:

The purpose of this study was to compare the occlusal parameters and the wear of artificial teeth between lingualized and anatomical occlusal complete dentures.

Methods

Edentulous patients who met the inclusion criteria were divided into three groups, and each patient made a complete denture. The Excellence IPN (IPN) group and Surpass G group were lingualized occlusal complete dentures, and Endura group was anatomical occlusal complete dentures. The occlusal contact area (OCA) and occlusal contact number (OCN) of the mandibular dentures was measured by silicone rubber translucency method, and the bite force was measured with Dental Prescale II at the 2-week, 3-month, and 12-month follow-up visits. The complete dentures were scanned at the 2-week and 12-month follow-up visits, and the wear of the maxillary and mandibular posterior teeth was assessed with software (Geomagic Qualify 2013; Raindrop). The data were analyzed by SPSS software.

Results

The results of One-way ANVOA showed that the OCA of the Endura group were significantly larger than the IPN and Surpass G groups at the 2-week follow-up visits, while the OCA of the IPN group were significantly larger than the Endura and Surpass G groups at the 3-month and 12-month follow-up visits ($P < 0.05$). The OCN of the three groups were significantly different at the 2-week, 3-month, and 12-month follow-up visits ($P < 0.05$). The bite force of IPN and Surpass G group were significantly lower than that of Endura group at the 2-week, 3-month, and 12-month follow-up visits ($P < 0.05$). The wear of IPN and Surpass G artificial teeth was significantly greater than Endura at the 12-month follow-up visits ($P < 0.05$).

Conclusions

The occlusal contact area increased as the complete dentures were used for longer periods of time. Compared with anatomical occlusion, lingualized occlusion was able to reduce the bite force. Wear varied among the three types of artificial teeth. (The Chinese Clinical Trial Registry (No. ChiCTR2300073420))

Background

The complete denture was an economical and widely used treatment for edentulous patients. Complete dentures relied on the sealing of the base edge to provide retention, recovered the oral function and

facial shape with the base and the artificial tooth. A stable occlusal relationship was also essential for the retention and stability of complete dentures [1, 2].

Studies had shown that compared with anatomical occlusion, lingualized occlusion could improve the retention stability of complete dentures among the patients with severe resorption in the medial alveolar ridge while enhanced the masticatory efficiency and quality of life [3]. Recent years, implant technology had provided new treatment options for edentulous jaws. However, the strict indications, the long treatment period and high costs could not meet the needs of most patients. Therefore, it was essential to carry out studies on the occlusion and clinical effects of complete dentures.

Occlusion was defined as the static relationship between the incising or occlusal surfaces of the maxillary or mandibular teeth or tooth analogues [4], which was closely related to oral health and considered as an important indicator of the functional status of the masticatory system [5]. The clinical effect of complete dentures could be affected directly by bite force and occlusal contact area (OCA). Therefore, the bite force and OCA were always the focus of the dentists during the treatment program [5, 6]. In clinical practice, articulating paper was a frequent approach for checking occlusion since it could roughly examine the quantity and position of occlusal contact. However, articulating paper could not quantify the OCA, and the operation procedure was highly subjective with low repeatability [7]. To examine the occlusion condition scientifically, quantified digital occlusal analysis instruments had been designed.

Digital occlusal analysis played important roles in the diagnosis, treatment and prognosis evaluation in the field of prosthodontics [5, 8, 9], orthodontics [10], implants, maxillofacial surgery [11], which could scientifically and objectively depict the restoration of oral function. The number, location, and size of occlusal contact sites, bite force, bite force distribution center, and balance of occlusal contact points were all assessed after the images were acquired and analyzed by the software [12]. The occlusal contact number (OCN) and OCA were quantitatively analyzed by the translucency of the silicone rubber material [5, 13, 14].

The wear of resin artificial teeth was a crucial factor in assessing the clinical performance of complete dentures [15]. Excessive wear of resin artificial teeth might result in reduced vertical distance, decreased chewing efficiency, and fatigue of the masticatory muscles of patients [16]. Currently, the majority of experimental procedures involved in vitro study or clinical observation. The in vitro studies frequently utilized two-body or three-body wear tests. However, the vitro device couldn't replicate the complex oral environment, which included the interplay of saliva, temperature, food, bite force, and the occlusal cycle on the wear of artificial teeth [17–19]. Clinical observation with digital technology could quantitatively analyze the morphological changes of artificial teeth and determine the wear condition of artificial teeth.

The purpose of this study was to evaluate the OCA, OCN, bite force, and wear of artificial teeth between the complete dentures with lingualized occlusion and anatomical occlusion. The null hypotheses of this study were that there were no differences between lingualized occlusal and anatomical occlusal

complete dentures in terms of OCA, OCN and bite force. After 12 months, there were no differences in the wear of the three types of artificial tooth.

Methods

Study design, IRB approval, and trial registry

A randomized single-blind parallel controlled clinical trial was conducted. The participants were divided into 3 groups randomly. Excellence IPN (Dentsply) group and Surpass G (GC) group were lingualized occlusal complete dentures (LOCD). Endura (Shofu) group arranged anatomical occlusal complete dentures (AOCD). The study was carried out in compliance with the Helsinki Declaration, approved by the Medical Ethics Committee of Dalian Stomatological Hospital (Ethics Review Batch No. DLKQLL2019015) and registered in the Chinese Clinical Trial Registry (No. ChiCTR2300073420). The selected patients signed the informed consent forms for the clinical trial.

Participants, sample size, randomization and dentures completed

Edentulous patients admitted to the Department of Prosthodontics of Dalian Stomatological Hospital from October 2019 to December 2022. The inclusion criteria were that edentulous patients whose remaining mandibular alveolar ridges were classified as classes II and III [20, 21], there were no symptoms of temporomandibular disorders, bruxism, xerostomia, orofacial motor disorders, severe oral manifestations of systematic disease or psychological/psychiatric conditions that could influence response to treatment, moderate thickness of alveolar ridge mucosa, were able to attend hospital appointments on a regular basis and assist in the collection of necessary data [3, 22, 23].

The sample size was calculated using G*Power software with selected statistical analyses of F-test and ANOVA–repeated measures, within factors. The following parameters were used: input effect size, $f = 0.25$; α error = 0.05, power = 0.80, number of groups = 3, and number of measurements = 3. The total sample size was 30, resulting in an approximate sample size = 10 per group [5, 24].

Randomization was performed using computer-generated numbers (Excel 2010, Microsoft). The number was put in a sealed envelope, and the participants ($n = 30$) were randomly allocated into one of three groups. Participants numbered 1–10 were assigned to the Endura group, 11–20 were assigned to the Excellence IPN group, and the remainder were assigned to the Surpass G group. One of the investigators, not involved in evaluating the treatment outcomes, generated the allocation sequence and assigned the participants to the groups [24].

New complete dentures were made according to standard procedures: (1) Obtaining the final impression and master casts by combined impression. (2) Recording jaw relationship and subsequent mounting the models in an average value articulator (Nissin). (3) Setting up denture tooth followed the specifications for lingualized occlusal and anatomical occlusal complete dentures. (4) The border molding was made

with Fit Checker (GC) after the wax dentures at final try-in. (5) After insertion and occlusal grinding of complete dentures, the patients were instructed to use and maintain the dentures, asked to return to the clinic after 2 weeks, 3 months and 12 months for follow-up visits, and visited the hospital in time for treatment if there was any pain or discomfort.

Measurements

The width and height of mandibular alveolar ridge were measured by the software (Geomagic Studio 2013; Raindrop), the mandibular models were scanned by scanner (X5; Sirona). The premolar area (between the first and second mandibular premolars) and the molar area (between the first and second mandibular molars) (Fig. 1A) were chosen based on anatomical features and position of the artificial tooth. The cropping and distance functions in the software were used to measure the height and width of the left and right alveolar ridge in the premolar area and molar area (Fig. 1B). All data were measured three times and averaged [25].

The OCA and OCN were measured by registering a check-bite in the intercuspatal position using silicone impression materials (Blue Silicone; GC). A tooth-contact analyzing device (Bite Eye; GC) was used to take measurements after setting thresholds such that occlusal contact thickness was 100 μm [5, 13, 14].

The bite force was measured by the Dental Prescale II (GC). The Dental Prescale II was rotated into the patients' mouth after they were selected based on the size of the dental arch. The patients were guided to hold by maximum force for at least 3 seconds. The Dental Prescale II was inserted into the scanner (Bite Force Analyzer; GC) to collect data [6, 12, 26].

The wear of artificial teeth was measured by software (Geomagic Qualify 2013; Raindrop). A scanner (X5; Sirona) was used to scan the maxillary and mandibular dentures at 2-week and 12-month follow-up visits. The data were imported into the Geomagic Qualify 2013. The digital model was cropped and only the artificial tooth portion was retained. The 2-week data was set as Reference, and the 12-month data were set as Test. The most severely abraded points of the buccal cusp, lingual cusp, and occlusal fossa were selected by data fitting and 3D analysis, vertical cross-sections were created at the location of these points (Fig. 2A), and the 2D comparison function was applied to measure the linear distance to obtain the vertical loss of artificial teeth (Fig. 2B). The same method was used to measure each part three times and the average value was calculated [27].

Statistical analysis

One-way ANOVA was used to analyze the premolar and molar height and width of mandibular alveolar ridges, bite force, OCA, OCN as well as the wear of artificial teeth.

Results

A total of 30 patients was included this study, and Table 1 displayed their ages, gender distributions, and alveolar bone state. There were no significant differences in the height and width at the mandibular

alveolar ridge premolar area or molar area among the three groups ($P > 0.05$) (Table 2).

The results of One-way ANVOA showed that the OCA of the Endura group ($12.08 \pm 2.30\text{mm}^2$) were significantly larger than the IPN ($7.35 \pm 1.89\text{mm}^2$) and Surpass G groups ($5.94 \pm 3.31\text{mm}^2$) at the 2-week follow-up visits, but the OCA of the IPN group ($24.42 \pm 5.72\text{mm}^2$, $33.61 \pm 6.31\text{mm}^2$) were significantly larger than the Endura ($14.29 \pm 2.50\text{mm}^2$, $19.41 \pm 3.95\text{mm}^2$) and Surpass G groups ($12.39 \pm 4.13\text{mm}^2$, $18.96 \pm 4.38\text{mm}^2$) at the 3-month and 12-month follow-up visits ($P < 0.05$) (Fig. 3). The OCN of the three groups were significantly different at the 2-week, 3-month, and 12-month follow-up visits ($P < 0.05$). The OCN of the IPN group (24.73 ± 5.48 , 31.36 ± 7.47) was significantly higher than the Surpass G group (16.40 ± 3.03 , 17.20 ± 4.08) at 3-month, and 12-month follow-up visits ($P < 0.05$) (Fig. 4).

The bite force of the IPN group ($328.04 \pm 71.01\text{N}$, $370.19 \pm 48.46\text{N}$, $397.13 \pm 54.43\text{N}$) and Surpass G group ($327.17 \pm 42.17\text{N}$, $347.33 \pm 53.07\text{N}$, $382.26 \pm 52.85\text{N}$) was significantly lower than that of the Endura group ($422.90 \pm 91.36\text{N}$, $455.78 \pm 104.79\text{N}$, $493.42 \pm 110.63\text{N}$) at the 2-week, 3-month, and 12-month follow-up visits, respectively ($P < 0.05$). There was no significant difference in the bite force between the IPN and Surpass G group at the 2-week, 3-month, and 12-month follow-up visits ($P > 0.05$) (Fig. 5).

The average wear of the IPN ($121.8 \pm 6.9\mu\text{m}$) and Surpass G ($94.0 \pm 3.8\mu\text{m}$) artificial teeth was significantly greater than Endura ($75.8 \pm 3.5\mu\text{m}$) at the 12-month follow-up visits ($P < 0.05$) (Fig. 6). The specific wear of artificial teeth was showed in Table 3.

Table 1
The distribution of edentulous patients among the three groups

	Cases (n)	Age	Male	Female	Class Ⅰ	Class Ⅱ
IPN	10	68	5	5	4	6
Surpass G	10	67	6	4	4	6
Endura	10	66	5	5	5	5

Table 2
Results of one-way ANOVA test for mandibular ridge height and width
(mm, $\bar{x} \pm s$)

	Premolar Area		Molar Area	
	Height	Width	Height	Width
IPN	6.71 $\pm$ 1.91	12.38 $\pm$ 2.66	6.56 $\pm$ 1.97	13.70 $\pm$ 2.62
Surpass G	5.47 $\pm$ 3.02	11.19 $\pm$ 2.57	5.65 $\pm$ 3.10	12.34 $\pm$ 3.32
Endura	7.11 $\pm$ 2.42	12.43 $\pm$ 1.70	6.99 $\pm$ 2.17	13.81 $\pm$ 1.10
F	1.169	0.857	0.759	1.009
<i>p</i>	0.326	0.436	0.478	0.378

* indicated significant differences in the comparison of one-way ANOVA tests between groups. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$.

Table 3
The wear of single artificial teeth ($\mu\text{m} \times \pm s$)

	Buccal Cusp			Occlusal Fossa			Lingual cusp		
	IPN	Surpass G	Endura	IPN	Surpass G	Endura	IPN	Surpass G	Endura
14	71.18 $\pm$ 55.81	34.29 $\pm$ 28.90	90.57 $\pm$ 24.61	61.41 $\pm$ 53.38	24.21 $\pm$ 19.93	83.46 $\pm$ 26.78	166.14 $\pm$ 59.20	132.56 $\pm$ 41.74	105.81 $\pm$ 35.13
15	81.38 $\pm$ 55.43	44.94 $\pm$ 29.58	95.94 $\pm$ 38.43	69.18 $\pm$ 54.92	29.58 $\pm$ 20.18	88.81 $\pm$ 28.28	184.40 $\pm$ 54.32	146.03 $\pm$ 45.57	112.36 $\pm$ 44.97
16	88.42 $\pm$ 66.82	48.61 $\pm$ 28.94	105.56 $\pm$ 36.01	75.26 $\pm$ 55.87	35.58 $\pm$ 22.46	99.17 $\pm$ 26.78	195.72 $\pm$ 62.08	158.48 $\pm$ 48.83	113.39 $\pm$ 21.28
17	63.18 $\pm$ 59.37	44.46 $\pm$ 33.57	97.31 $\pm$ 33.06	62.85 $\pm$ 53.61	29.55 $\pm$ 19.35	78.35 $\pm$ 16.22	169.49 $\pm$ 52.21	132.92 $\pm$ 44.48	96.02 $\pm$ 19.75
24	69.28 $\pm$ 58.97	31.33 $\pm$ 25.25	90.84 $\pm$ 25.42	61.59 $\pm$ 52.88	25.49 $\pm$ 20.91	81.00 $\pm$ 30.93	165.14 $\pm$ 61.43	119.17 $\pm$ 49.71	106.62 $\pm$ 32.99
25	82.03 $\pm$ 58.98	41.97 $\pm$ 25.44	99.69 $\pm$ 34.28	68.75 $\pm$ 52.78	33.09 $\pm$ 20.29	89.61 $\pm$ 30.26	189.72 $\pm$ 52.91	145.76 $\pm$ 42.85	113.18 $\pm$ 44.31
26	83.39 $\pm$ 63.01	49.93 $\pm$ 24.31	107.20 $\pm$ 34.32	79.78 $\pm$ 50.55	37.03 $\pm$ 19.09	101.58 $\pm$ 30.13	201.06 $\pm$ 71.31	158.18 $\pm$ 53.07	116.64 $\pm$ 26.40
27	65.08 $\pm$ 64.11	30.26 $\pm$ 25.05	98.18 $\pm$ 33.11	60.30 $\pm$ 55.65	26.21 $\pm$ 15.90	79.00 $\pm$ 24.73	170.14 $\pm$ 50.70	131.29 $\pm$ 44.82	96.29 $\pm$ 25.92
34	64.40 $\pm$ 42.29	30.96 $\pm$ 28.14	90.78 $\pm$ 39.79	167.41 $\pm$ 52.16	117.52 $\pm$ 47.62	92.16 $\pm$ 47.58	68.40 $\pm$ 35.41	28.78 $\pm$ 23.45	90.36 $\pm$ 34.77
35	83.98 $\pm$ 73.42	49.54 $\pm$ 36.09	93.87 $\pm$ 29.82	190.28 $\pm$ 53.26	142.83 $\pm$ 48.24	114.32 $\pm$ 40.86	78.14 $\pm$ 30.52	50.21 $\pm$ 40.05	98.41 $\pm$ 72.71
36	95.23 $\pm$ 80.21	61.48 $\pm$ 47.82	107.04 $\pm$ 34.52	206.82 $\pm$ 52.35	166.55 $\pm$ 52.06	121.29 $\pm$ 75.22	84.48 $\pm$ 39.47	57.84 $\pm$ 43.65	103.22 $\pm$ 34.59
37	68.79 $\pm$ 53.20	47.57 $\pm$ 36.94	96.14 $\pm$ 20.38	172.19 $\pm$ 54.62	138.61 $\pm$ 45.01	95.70 $\pm$ 35.13	72.54 $\pm$ 32.26	52.34 $\pm$ 39.84	98.89 $\pm$ 37.19
44	66.91 $\pm$	33.68 $\pm$ 40.76	90.01 $\pm$ 33.13	167.67 $\pm$ 51.82	127.34 $\pm$ 43.76	96.36 $\pm$ 40.15	72.72 $\pm$ 33.07	43.70 $\pm$ 47.02	92.91 $\pm$ 56.14

	Buccal Cusp		Occlusal Fossa				Lingual cusp		
	43.09								
45	83.85 ± 66.74	46.54 ± 38.91	101.59 ± 34.86	187.73 ± 54.71	142.08 ± 47.98	113.88 ± 52.81	75.15 ± 35.85	49.11 ± 28.67	97.28 ± 61.42
46	99.03 ± 84.36	44.88 ± 34.52	105.73 ± 59.27	199.68 ± 57.46	148.88 ± 50.99	113.19 ± 27.84	86.55 ± 31.63	61.41 ± 38.77	106.89 ± 43.98
47	66.49 ± 58.07	42.08 ± 26.53	97.81 ± 37.40	169.98 ± 50.56	128.29 ± 46.02	94.15 ± 29.73	76.33 ± 34.27	48.38 ± 31.19	96.58 ± 23.81

Discussion

The main findings of this study were the significant differences in the OCA, OCN, bite force, and the wear of artificial teeth among the three groups. Consequently, the null hypothesis had to be rejected.

The height and width of alveolar ridge in the mandibular premolar and molar areas were measured by using the software in this study. As the condition of the alveolar ridge had a substantial impact on the restorative effect of complete dentures. Statistical analysis showed that there was no significant difference in the height and width of the alveolar ridge among the three groups (Table 1). Moreover, this study selected complete dentures worn for two weeks as the baseline to measure the occlusal parameter and wear of artificial tooth, since clinical adjustment for the pain or discomfort of denture had been done at that time.

Occlusion was crucial to the process of mastication. The physiologic state of occlusal contact had an impact on both the masticatory cycle and occlusal health. LOCD was significant smaller in OCA and OCN than AOCD after 2-week follow-up visits. The reason was that the LOCD buccal cusps of the maxillary posterior teeth did not contact with the opposite jaw, only the lingual cusps did. The OCA was not substantially different between the IPN and Surpass G groups, but the OCN was significantly different between the IPN and Surpass G groups. This might be due to differences in the design of the cusp inclination and occlusal contact position of the artificial tooth. At the 3-month follow-up visits, OCA increased and OCN decreased in all three groups. The reason could be that the cusps with occlusal contacts became smooth, and neighboring small occlusal points were worn out to form a large occlusal point during the 3 months use of complete denture. The OCA of the mandibular dentures increased further at the 12-month follow-up visit, and the OCN of the lingualized occlusal complete dentures increased as well. The occlusal contact of artificial tooth changed from point to plane contact, resulting in a bigger occlusal contact area after 12 months. The increased OCN of LOCD was owing to the buccal cusps of the premolars and molars contacted.

The bite force, which was investigated in the field of oral biomechanics, was the most significant indicator of masticatory system health. Maximum bite force reflected the masticatory abilities, which could give objective data for evaluating the occlusal system, joints, and masticatory muscles [28]. Many factors, including age, health, facial shape, masticatory muscle strength, and tooth contact area, impacted the amount of bite force [29, 30]. The results of this study confirmed that bite force increased as denture wearing time increased [31, 32]. The bite force of LOCD was found to be significantly lower than that of AOCD at the 2-week, 3-month, and 12-month follow-up visits because of lateral force minimized in LOCD.

In this study, a digital method was used to quantify and analyze the wear of artificial teeth. The average wear of the IPN ($121.75 \pm 6.56\mu\text{m}$) and Surpass G ($93.96 \pm 3.60\mu\text{m}$) artificial teeth was significantly greater than Endura ($75.78 \pm 3.26\mu\text{m}$) after 12 months. These results were similar with the results of previous clinical studies. Schmid-Schwap et al [33] measured median vertical loss for posterior denture teeth of 121–221 μm after 1 year of clinical service. Stober et al [17] found that the vertical loss from composite resin denture teeth for complete denture wearers increased continuously over 12 months up to maximum vertical loss of denture teeth of 146 (148) μm at 12 months. Artificial teeth wear varied depending on hardness, bite force and diet [33, 34]. It was worth noting that the artificial posterior teeth wear areas were different in the maxillary and mandibular dentures. The lingual inclined planes of the maxillary buccal cusps, the buccal and lingual inclined planes of the mandibular buccal cusps, and the buccal inclined planes of the mandibular lingual cusps exhibited varying degrees of wear between the IPN and Surpass G group. The reason was the friction between foods and artificial teeth, as well as the direct contact of the artificial teeth. The lateral movement caused wear on the lingual inclined planes and buccal inclined planes of the mandibular buccal cusps.

The limitations of this study were that the sample size was smaller and only the classification of alveolar ridge was considered. Although the statistically significant differences in the occlusal parameters and wear of artificial teeth among tested groups were found in this study, further studies should increase the number of cases according to the inclusion criteria and be conducted with stratified randomized design based on age, gender, dietary habits, etc., to better evaluate the differences between lingualized and anatomical occlusal complete dentures.

Conclusions

The occlusal contact area increased as the complete dentures were used for 12 months. Compared with anatomical occlusion, lingualized occlusion was able to reduce the bite force of complete dentures. Wear varied among the three types of artificial teeth.

Abbreviations

IPN

Excellence IPN

OCA
occlusal contact area
OCN
occlusal contact number
LOCD
lingualized occlusal complete dentures
AOCD
anatomical occlusal complete dentures

Declarations

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Authorship Contribution Statement

ZQW: Participate in the design and development of research protocols, implementation of the research process, collation of data, statistical analysis, drafting and revision of papers. **YS:** Participate in data counting, checking, analyzing, and revising papers. **JMW:** fabricate the complete dentures **YL:** Participate in the design and development of the research program and guide to carry out the implementation. **WZX:** Participate in proposing research topics, designing and developing research proposals, mentoring support, researching and organizing literature, revising papers.

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Availability of data and materials

Data that support the findings of this study are available from the corresponding author, upon reasonable request.

Ethics approval and consent to participate

The study was carried out in compliance with the Helsinki Declaration, approved by the Medical Ethics Committee of Dalian Stomatological Hospital (Ethics Review Batch No. DLKQLL2019015) and registered in the Chinese Clinical Trial Registry (No. ChiCTR2300073420). Informed consent was obtained from all subjects involved in the study.

Consent for publication

Not applicable.

Competing interests

The authors declare no conflict of interest.

References

1. Bozhkova T, Musurlieva N, Slavchev D et al. Occlusal Indicators Used in Dental Practice: A Survey Study[J]. *Biomed Res Int*, 2021, 2021: 2177385. <https://doi.org/10.1155/2021/2177385>.
2. Goldstein G, Kapadia Y, Campbell S. Complete Denture Occlusion: Best Evidence Consensus Statement[J]. *J Prosthodont*, 2021, 30(S1): 72 – 7. <https://doi.org/10.1111/jopr.13309>.
3. Kawai Y, Ikeguchi N, Suzuki A et al. A double blind randomized clinical trial comparing lingualized and fully bilateral balanced posterior occlusion for conventional complete dentures[J]. *J Prosthodont Res*, 2017, 61(2): 113 – 22. <https://doi.org/10.1016/j.jpor.2016.07.003>.
4. The Glossary of Prosthodontic Terms: Ninth Edition[J], *Prosthet Dent J*. 2017, 117(5S): e63. <https://doi.org/10.1016/j.prosdent.2016.12.001>.
5. Lee HS, Ko KH, Huh YH et al. Correlation between occlusal contact area at various levels of interocclusal thicknesses and masticatory performance[J]. *J Oral Rehabil*, 2022, 49(5): 522-8. <https://doi.org/10.1111/joor.13292>.
6. Zhao Z, Wang Q, Li J et al. Construction of a novel digital method for quantitative analysis of occlusal contact and force[J]. *BMC Oral Health*, 2023, 23(1): 190. <https://doi.org/10.1186/s12903-023-02899-y>.
7. Qadeer S, Ozcan M, Edelhoff D et al. Accuracy, Reliability and Clinical Implications of Static Compared to Quantifiable Occlusal Indicators[J]. *Eur J Prosthodont Restor Dent*, 2021, 29(3): 130 – 41. https://doi.org/10.1922/EJPRD_2202Qadeer12.
8. Idris RI, Shoji Y, Lim TW. Occlusal force and occlusal contact reestablishment with resin-bonded fixed partial dental prostheses using the Dahl concept: A clinical study[J]. *J Prosthet Dent*, 2022, 127(5): 737 – 43. <https://doi.org/10.1016/j.prosdent.2020.11.035>.
9. Wada M, Mameno T, Tsujioka Y et al. Effective utilization of digital technology in complete denture fabrication[J]. *J Oral Sci*, 2022, 64(2): 172-4. <https://doi.org/10.2334/josnurd.21-0381>.
10. Pinheiro F, Frota CM, Garib DG et al. A Cleft-Customized Occlusal Rating System to Assess Orthodontic Occlusal Improvement in Patients With Unilateral Cleft Lip and Palate[J]. *Cleft Palate Craniofac J*, 2022, 59(1): 54–65. <https://doi.org/10.1177/1055665621995313>.
11. Kang YF, Ding MK, Qiu SY et al. Mandibular Reconstruction Using Iliac Flap Based on Occlusion-Driven Workflow Transferred by Digital Surgical Guides[J]. *J Oral Maxillofac Surg*, 2022, 80(11): 1858-65. <https://doi.org/10.1016/j.joms.2022.07.140>.
12. Shiga H, Komino M, Uesugi H et al. Comparison of two dental prescale systems used for the measurement of occlusal force[J]. *Odontology*, 2020, 108(4): 676 – 80. <https://doi.org/10.1007/s10266-020-00509-9>.

13. Imamura Y, Sato Y, Kitagawa N et al. Influence of occlusal loading force on occlusal contacts in natural dentition[J]. *J Prosthodont Res*, 2015, 59(2): 113 – 20.
<https://doi.org/10.1016/j.jpor.2014.07.001>.
14. Horie T, Kanazawa M, Komagamine Y et al. Association between near occlusal contact areas and mixing ability[J]. *J Oral Rehabil*, 2014, 41(11): 829 – 35. <https://doi.org/10.1111/joor.12216>.
15. Stober T, Henninger M, Schmitter M et al. Three-body wear of resin denture teeth with and without nanofillers[J]. *J Prosthet Dent*, 2010, 103(2): 108 – 17. [https://doi.org/10.1016/S0022-3913\(10\)60014-5](https://doi.org/10.1016/S0022-3913(10)60014-5).
16. Pham DM, Gonzalez MD, Ontiveros JC et al. Wear Resistance of 3D Printed and Prefabricated Denture Teeth Opposing Zirconia[J]. *J Prosthodont*, 2021, 30(9): 804 – 10.
<https://doi.org/10.1111/jopr.13339>.
17. Stober T, Geiger A, Rues S et al. Factors affecting wear of composite resin denture teeth–24-month results from a clinical study[J]. *Clin Oral Investig*, 2012, 16(2): 413 – 20.
<https://doi.org/10.1007/s00784-011-0534-y>.
18. Heintze SD, Rousson V, Stober T. Patient- and therapy-related factors on the wear of denture teeth–results of a clinical trial[J]. *Dent Mater*, 2015, 31(3): 302-7.
<https://doi.org/10.1016/j.dental.2014.12.009>.
19. Jin S, Choi JW, Jeong CM et al. Evaluating the Wear of Resin Teeth by Different Opposing Restorative Materials[J]. *Materials (Basel)*, 2019, 12(22): 3684. <https://doi.org/10.3390/ma12223684>.
20. Atwood DA. Reduction of residual ridge: A major oral disease entity[J]. *J Prosthet Dent*, 1971, 26(3): 266 – 79. [https://doi.org/10.1016/0022-3913\(71\)90069-2](https://doi.org/10.1016/0022-3913(71)90069-2).
21. Cawood JI, Howell RA. A classification of the edentulous jaws[J]. *Int J Oral Maxillofac Surg*, 1988, 17(4): 232-6. [https://doi.org/10.1016/s0901-5027\(88\)80047-x](https://doi.org/10.1016/s0901-5027(88)80047-x).
22. Kimoto S, Gunji A, Yamakawa A et al. Prospective clinical trial comparing lingualized occlusion to bilateral balanced occlusion in complete dentures: a pilot study[J]. *Int J Prosthodont*, 2006, 19(1): 103-9. <https://doi.org/>.
23. Matsumaru Y. Influence of mandibular residual ridge resorption on objective masticatory measures of lingualized and fully bilateral balanced denture articulation[J]. *J Prosthodont Res*, 2010, 54(3): 112-8. <https://doi.org/10.1016/j.jpor.2009.11.008>.
24. Limpuangthip N, Tumrasvin W, Pinyowattanasilp T et al. Evaluation of dynamic impression lining material effects on masticatory function and the oral health-related quality of life of complete denture wearers: A six-month randomized controlled trial[J]. *J Prosthodont Res*, 2022.
https://doi.org/10.2186/jpr.JPR_D_22_00004.
25. Xia Y, Ma C, Chen J et al. Mandibular residual ridge morphology in relation to complete dentures and implant overdentures-Part I: Predictors for perceived conventional denture stability[J]. *Clin Implant Dent Relat Res*, 2021, 23(1): 131-9. <https://doi.org/10.1111/cid.12976>.
26. Baca E, Yengin E, Gokcen-Rohlig B et al. In vivo evaluation of occlusal contact area and maximum bite force in patients with various types of implant-supported prostheses[J]. *Acta Odontol Scand*,

- 2013, 71(5): 1181-7. <https://doi.org/10.3109/00016357.2012.757360>.
27. Kühne C, Lohbauer U, Raith S et al. Measurement of Tooth Wear by Means of Digital Impressions: An In-Vitro Evaluation of Three Intraoral Scanning Systems[J]. Appl Sci, 2021, 11(11): 5161. <https://doi.org/>.
28. Sonogo MV, Goiato MC, Dos Santos DM. Electromyography evaluation of masseter and temporalis, bite force, and quality of life in elderly patients during the adaptation of mandibular implant-supported overdentures[J]. Clin Oral Implants Res, 2017, 28(10): e169-e74. <https://doi.org/10.1111/clr.12980>.
29. Thongudomporn U, Chongsuvivatwong V, Geater AF. The effect of maximum bite force on alveolar bone morphology[J]. Orthod Craniofac Res, 2009, 12(1): 1–8. <https://doi.org/10.1111/j.1601-6343.2008.01430.x>.
30. Koc D, Dogan A, Bek B. Bite force and influential factors on bite force measurements: a literature review[J]. Eur J Dent, 2010, 4(2): 223 – 32. <https://doi.org/10.1055/s-0039-1697833>.
31. Fayad MI, Alruwaili HHT, Khan MS et al. Bite Force Evaluation in Complete Denture Wearer with Different Denture Base Materials: A Randomized Controlled Clinical Trial[J]. J Int Soc Prev Community Dent, 2018, 8(5): 416-9. https://doi.org/10.4103/jispcd.JISPCD_2_18.
32. Shala K, Tmava-Dragusha A, Dula L et al. Evaluation of Maximum Bite Force in Patients with Complete Dentures[J]. Open Access Maced J Med Sci, 2018, 6(3): 559 – 63. <https://doi.org/10.3889/oamjms.2018.141>.
33. Schmid-Schwab M, Rousson V, Vornwagner K et al. Wear of two artificial tooth materials in vivo: a 12-month pilot study[J]. J Prosthet Dent, 2009, 102(2): 104 – 14. [https://doi.org/10.1016/S0022-3913\(09\)60118-9](https://doi.org/10.1016/S0022-3913(09)60118-9).
34. Zeng J, Sato Y, Ohkubo C et al. In vitro wear resistance of three types of composite resin denture teeth[J]. J Prosthet Dent, 2005, 94(5): 453-7. <https://doi.org/10.1016/j.prosdent.2005.08.010>.

Figures

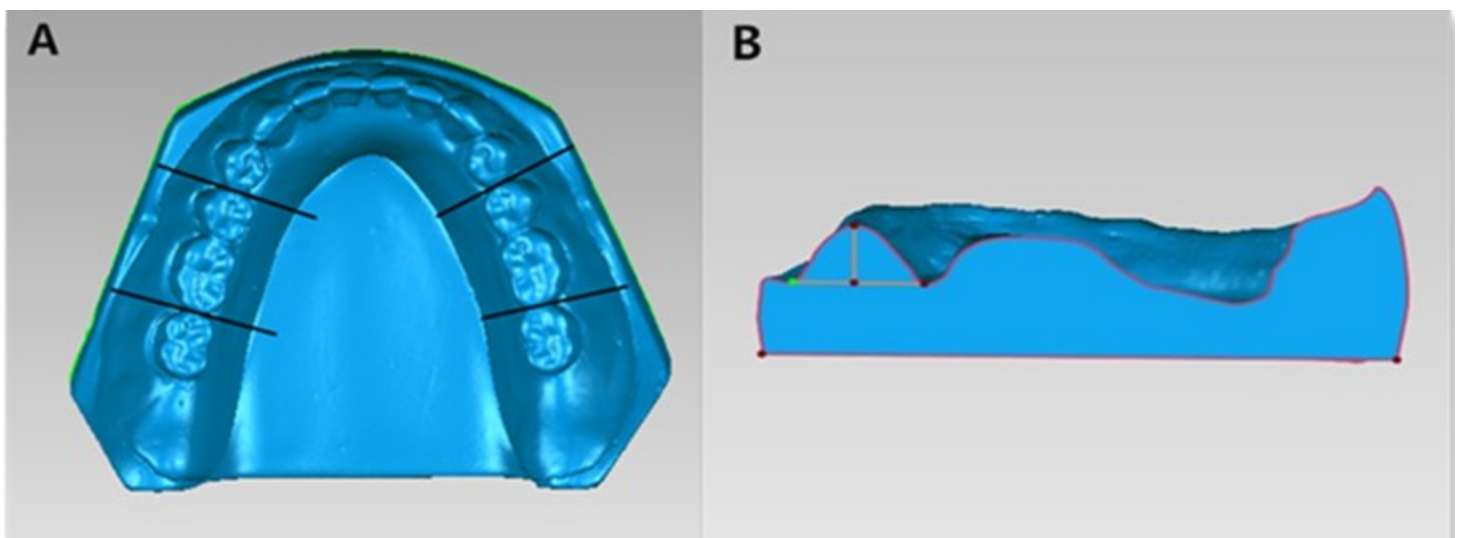


Figure 1

(A) Selection of alveolar ridge cross-sectional positions in the premolar and molar areas. (B) the width and height of mandibular alveolar ridge.

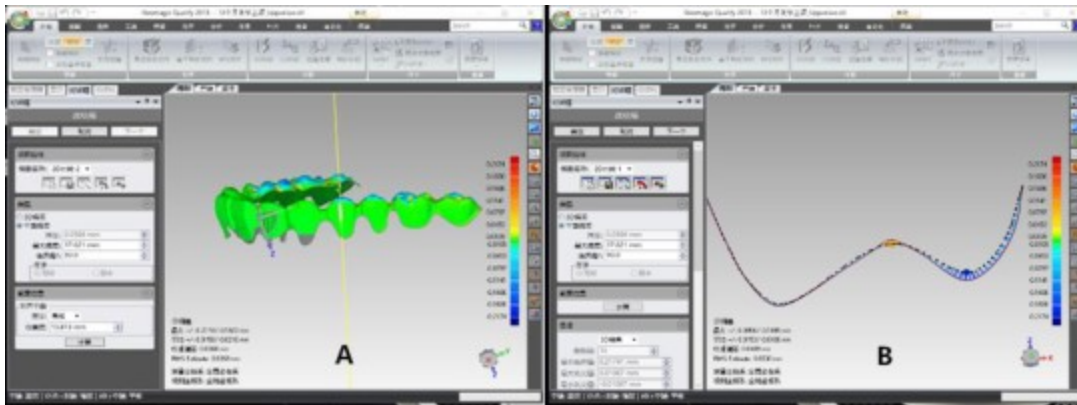


Figure 2

(A) create survey section. (B) comparative result.

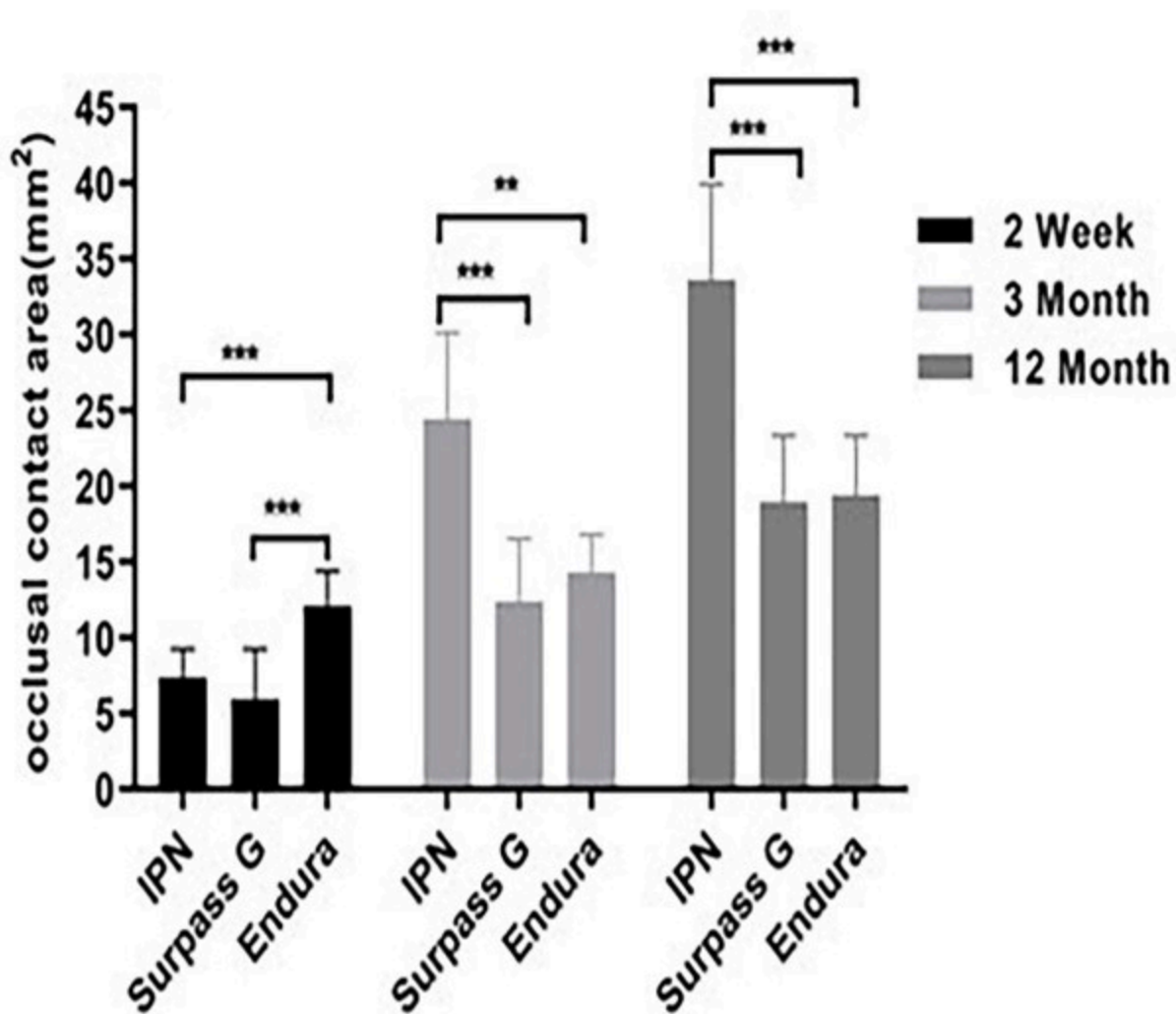


Figure 3

the occlusal contact area

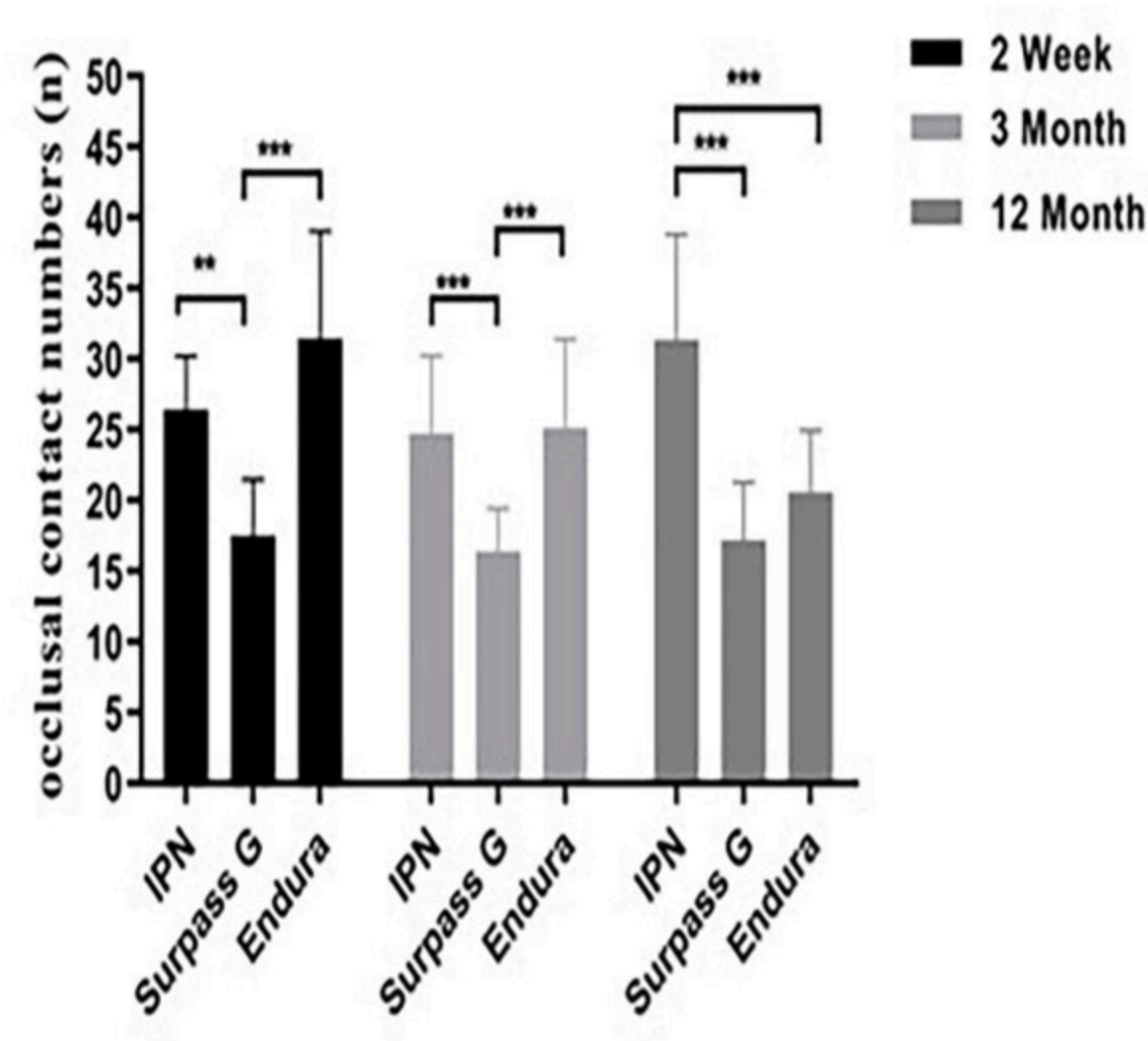


Figure 4

the occlusal contact numbers

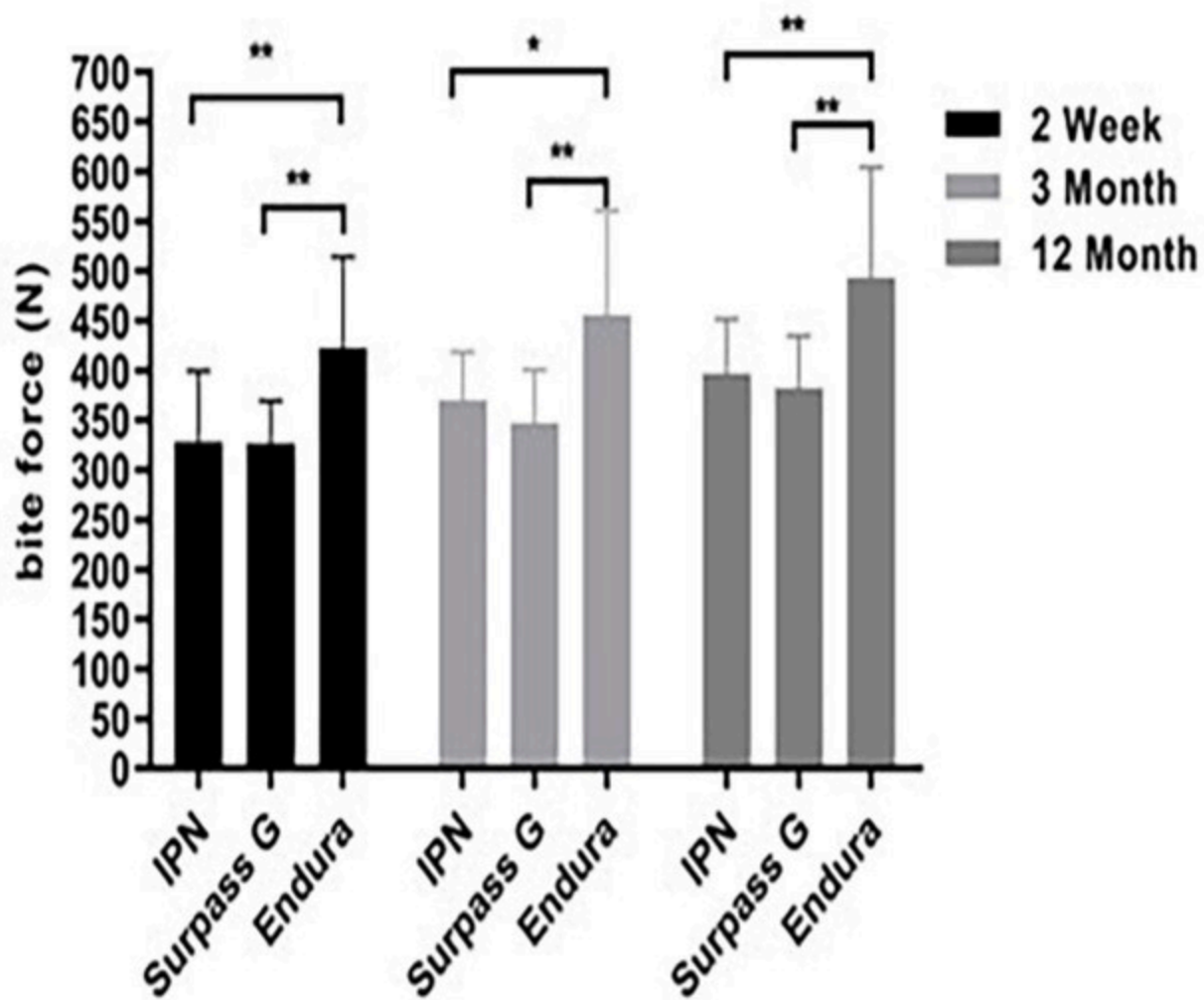


Figure 5

the bite force of three groups

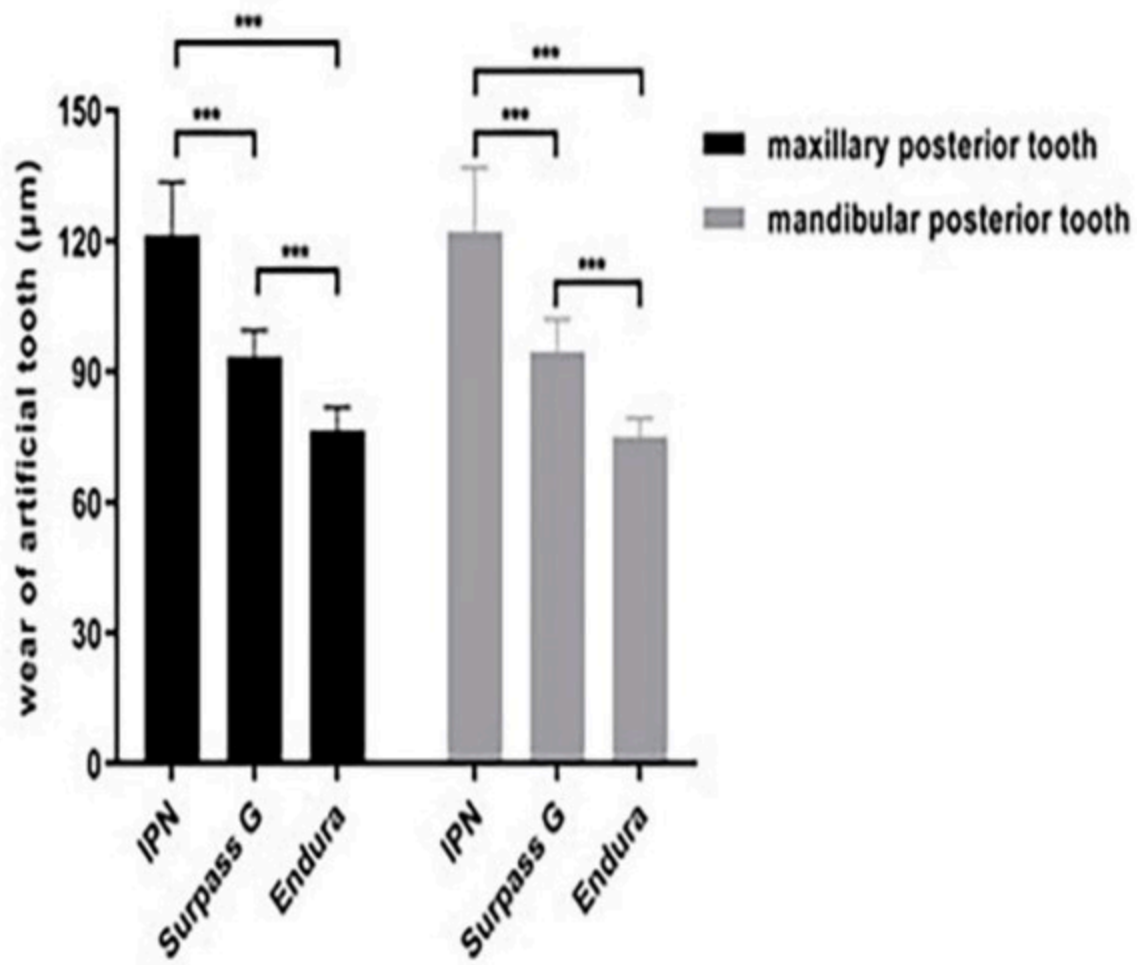


Figure 6

average wear of artificial tooth.